A QUASI-REGULAR IDEAL IS FINITELY GENERATED

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Quasi-regular ideal, defined through the quasi-regularity of the sequences generating it locally. defined in the same section, source lines 8100-8138, before the counted statement. Supplies the hypothesis and the local generators.
- Quasi-compactness of the closed subset defined by an ideal and the resulting finite cover by principal opens. used at proof lines 8144-8148. Produces the finitely many localising elements of the cover.
- A module or ideal generated by given elements on each piece of a finite cover is generated by them globally. the author cites Algebra, Lemma algebra-lemma-cover (tag 00EO) at proof lines 8155-8157. Concludes that the lifted elements together generate the ideal.

Source reference. Original lemma 07CW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9391–9391. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00EO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 542–542. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *A quasi-regular ideal is finitely generated.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $I \subset R$ be a quasi-regular ideal. Since $V(I)$ is quasi-compact, there exist $g_1, \ldots, g_m \in R$ such that $V(I) \subset D(g_1) \cup \ldots \cup D(g_m)$ and such that I_{g_j} is generated by a quasi-regular sequence $g_{j1}, \ldots, g_{jr_j} \in R_{g_j}$. Write $g_{ji} = g'_{ji}/g_j^{e_{ij}}$ for some $g'_{ij} \in I$. Write $1 + x = \sum g_j h_j$ for some $x \in I$ which is possible as $V(I) \subset D(g_1) \cup \ldots \cup D(g_m)$. Note that $\mathrm{Spec}(R) = D(g_1) \cup \ldots \cup D(g_m) \bigcup D(x)$ Then I is generated by the elements g'_{ij} and x as these generate on each of the pieces of the cover, see Algebra, Lemma 00EO. $\square$